Problem 1 (30 pts)

An electromagnetic surface wave propagates on a grounded dielectric slab as shown below. (The structure is infinite in the y direction.) In the air region above the slab ($z > 0$), the electric field in the phasor domain has the following form:

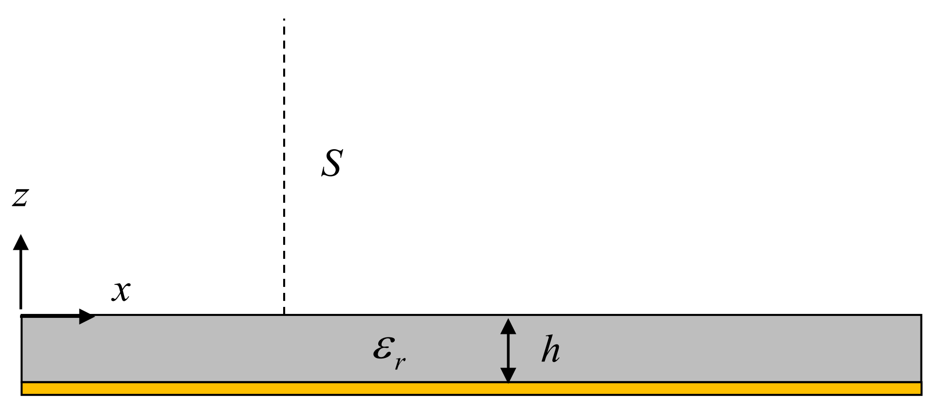
$$\underline{E}(x,z) = \underline{\hat{y}}\, e^{-jk_x x}\, e^{-\alpha_z z} \quad [\mathrm{V/m}],$$

where k_x and α_z are both positive real numbers. Note that there is no y variation of the fields of this surface wave.

a) Find the magnetic field vector $\underline{H}$ in the air region, in the phasor domain.

b) Find the complex Poynting vector in the air region.

c) Find the time average power in watts going (from left to right) through a surface S, which extends from the top of the slab ($z = 0$) to infinity in the vertical z direction, and is one meter wide in the y direction.



SOLUTION

Part (a)

$$\underline{H} = -\frac{1}{j\omega\mu_0}\nabla\times\underline{E}$$

so

$$\underline{H} = -\frac{1}{j\omega\mu_0}\left[\underline{\hat{x}}(\alpha_z) + \underline{\hat{z}}(-jk_x)\right]e^{-jk_x x}\, e^{-\alpha_z z} \quad [\mathrm{A/m}]$$

Part (b)

$$\underline{S} = \frac{1}{2}\underline{E}\times\underline{H}^*$$

$$\underline{S} = -\frac{1}{2\omega\mu_0}e^{-2\alpha_z z}\left[\underline{\hat{x}}(k_x) + j\underline{\hat{z}}(\alpha_z)\right] \quad \left[\mathrm{VA/m^2}\right]$$

Part (c)

$$P = \int_0^\infty \mathrm{Re}\,\underline{S}\cdot\underline{\hat{x}}\,dz$$

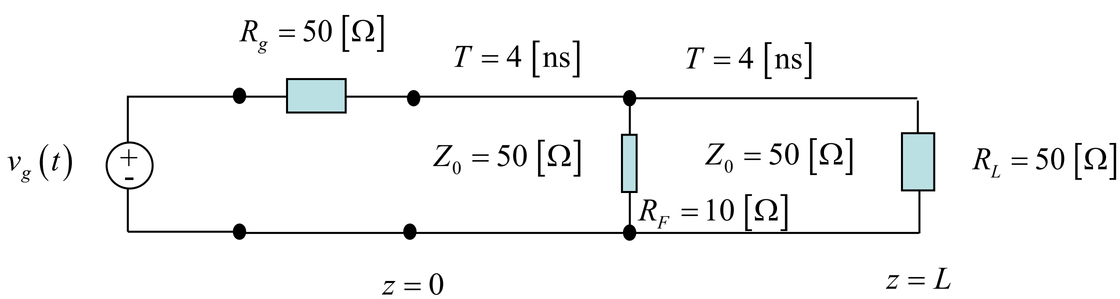
$$P = \frac{k_x}{4\omega\mu_0\alpha_z} \quad [\mathrm{W}]$$

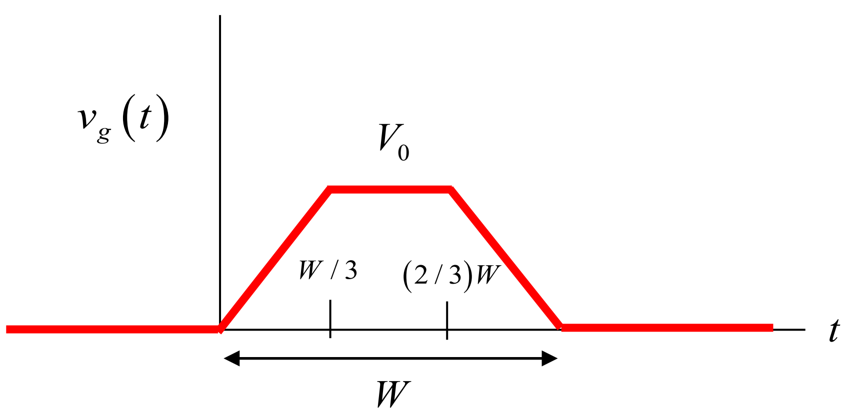
Problem 2 (35 pts)

A voltage source is applied at the left end of a transmission line as shown below. The transmission line meets a second transmission line, which is then terminated by a load. At the junction between the two transmission lines a fault occurs, which is modeled by a parallel (shunt) resistor R_F.

A plot of the generator voltage $v_g(t)$ is shown below. The pulse peak is $V_0 = 8\,[\mathrm{V}]$ and the width of the pulse is $W = 3\,[\mathrm{ns}]$.

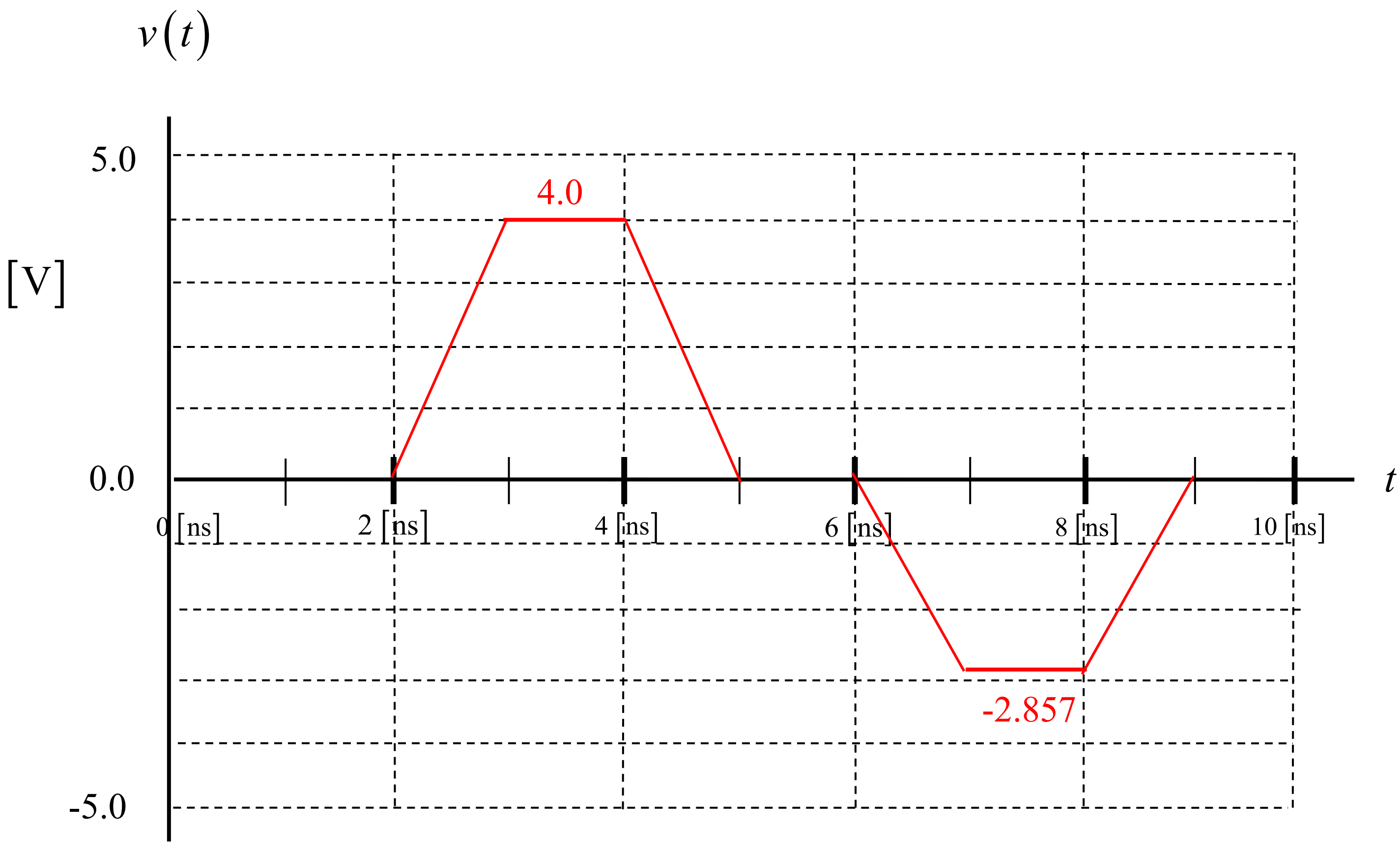
Plot the voltage $v(t)$ measured by an oscilloscope that is connected to the first (left) line at a point halfway down the first line (halfway between the generator and the fault). Plot to a time of 10 [ns]. Use the graph on the next page to make your plot. Label all voltage values on your plot.





Make your plot here:

The plot is shown below. The calculation is done on the next page.



SOLUTION

The load seen by the first transmission line is 50 [Ω] in parallel with 10 [Ω], or 8.3333 [Ω]. The load reflection coefficient is then

$$\Gamma_J^+ = -0.71429 .$$

We also have

$$\Gamma_g = 0$$

and

$$A = \frac{1}{2} .$$

We then use

$$\begin{aligned} v(z,t) = 0.5v_g&\left(t - z/c_d\right) \\ &+\Gamma_J^+ A v_g\left(t - L/c_d - (L-z)/c_d\right) \\ &\quad+\cancel{\Gamma}_g\Gamma_J^+ A v_g\left(t - 2L/c_d - z/c_d\right) \\ &\qquad+\cancel{\Gamma}_g\Gamma_J^{+2} A v_g\left(t - 3L/c_d - (L-z)/c_d\right) \\ &\qquad\quad+\ldots \end{aligned}$$

This gives us

$$\begin{aligned} v(z,t) = 0.5v_g&\left(t - 2.0[\text{nS}]\right) \\ &+(-0.71429)(0.5)v_g\left(t - 4.0[\text{nS}] - 2.0[\text{nS}]\right) \end{aligned}$$

or

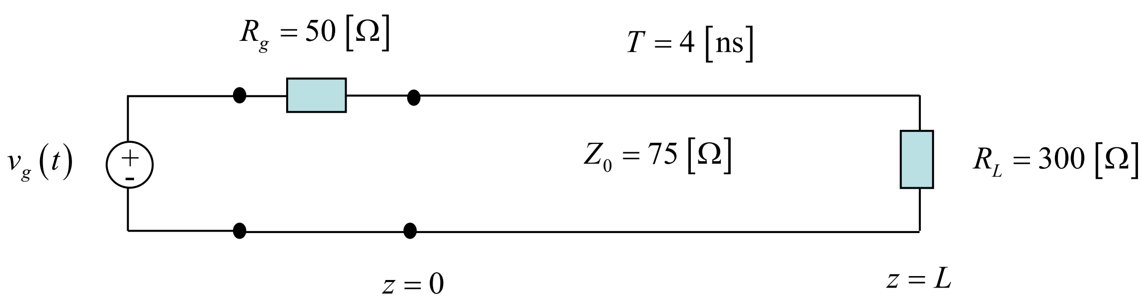
$$v(z,t) = 0.5v_g\left(t - 2.0[\text{nS}]\right) - 0.35714v_g\left(t - 6.0[\text{nS}]\right)$$

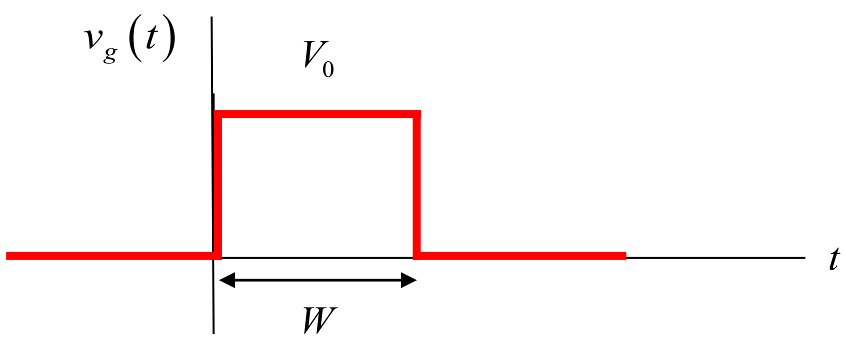
Problem 3 (35 pts)

A voltage source is applied at the left end of a transmission line as shown below. A plot of the generator voltage is shown below. The pulse width is $W = 6\,[\text{ns}]$ and the pulse voltage is $V_0 = 5\,[\text{V}]$.

a) Construct a bounce diagram for this problem that extends to a time of 16 [ns]. (Make your bounce diagram on the next page.)

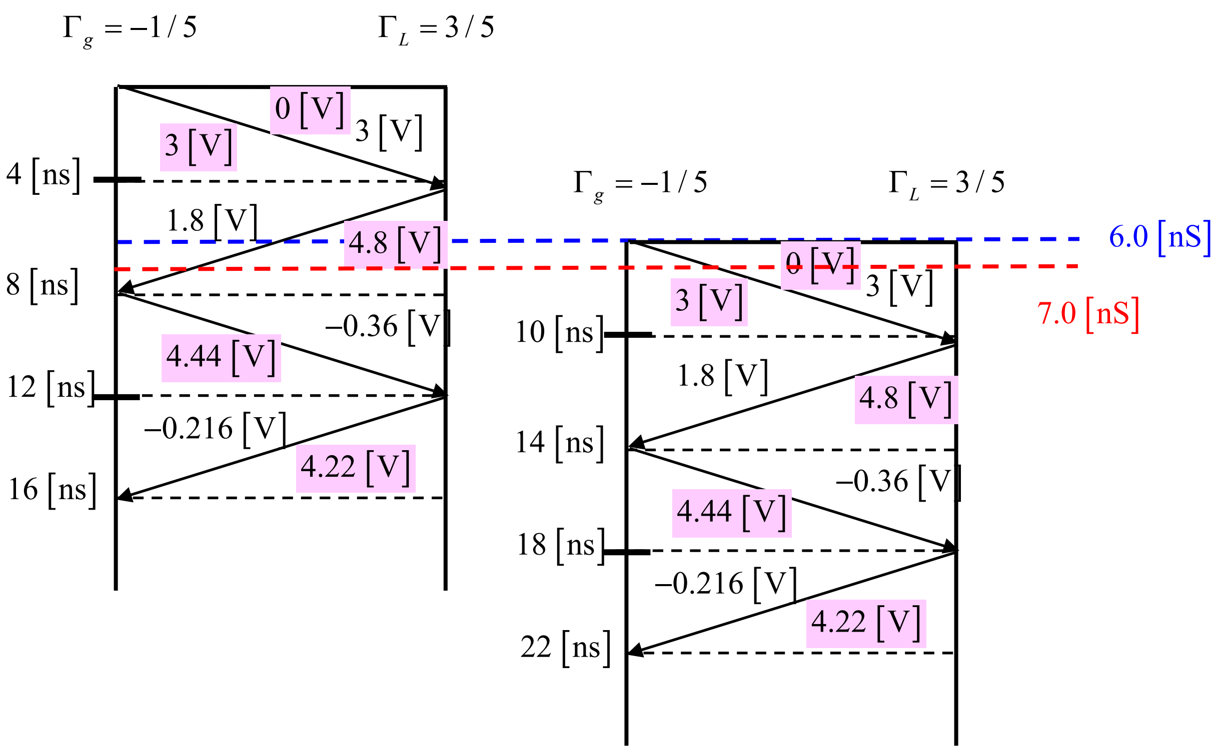
b) Make an accurate "snapshot" plot of the voltage $v(z)$ on the line at $t = 7.0\,[\text{ns}]$. Make your plot on the graph that is given below on the next page. Label all voltage values on your plot.



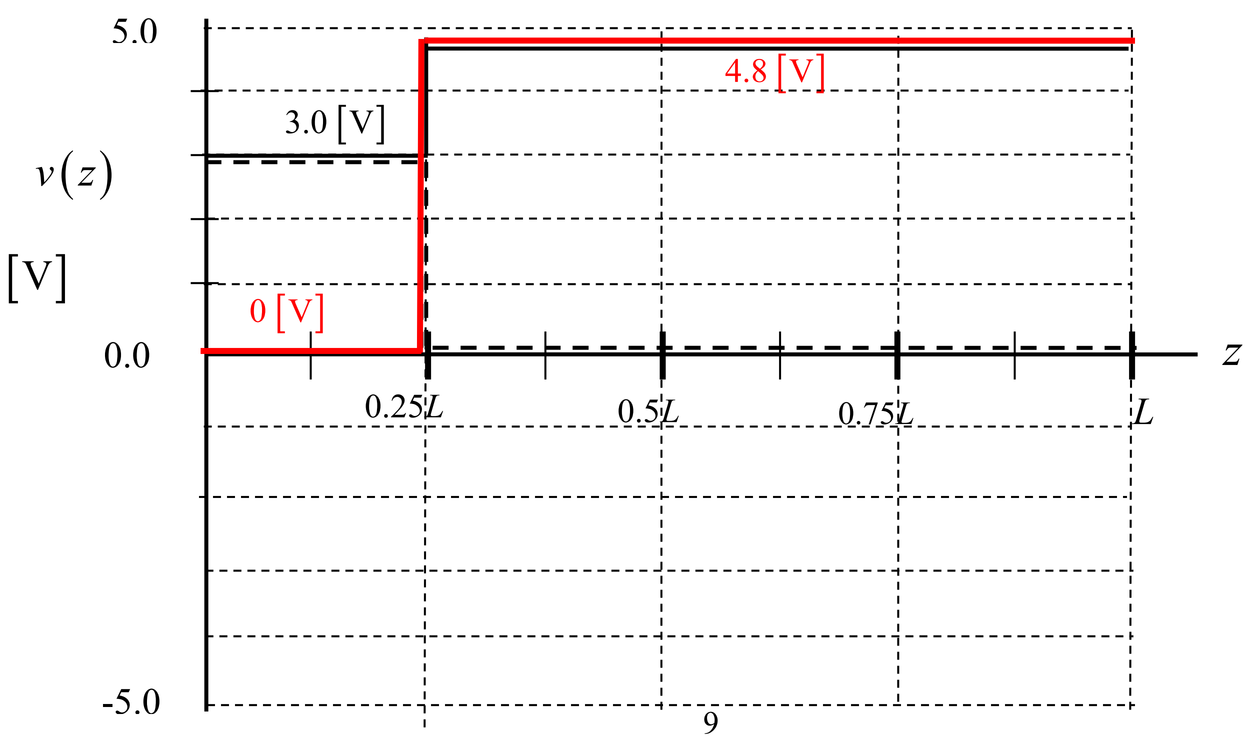


SOLUTION

The original bounce diagram and the shifted one are shown below.



The snapshot due to the original bounce diagram (black solid line) and the shifted bounce diagram (black dashed line) are shown below. The final plot (coming from the black solid plot minus the black dashed plot) is shown in red.



Problem 1 (30 pts)

Fields exists inside of a coaxial type of region that is described in cylindrical coordinates. The fields exists in the coaxial region $a < \rho < b$, $0 < \phi < 2\pi$, $0 < z < L$. This region is nonmagnetic ($\mu = \mu_0$) having a relative permittivity ε_r. The magnetic field in this region is given by

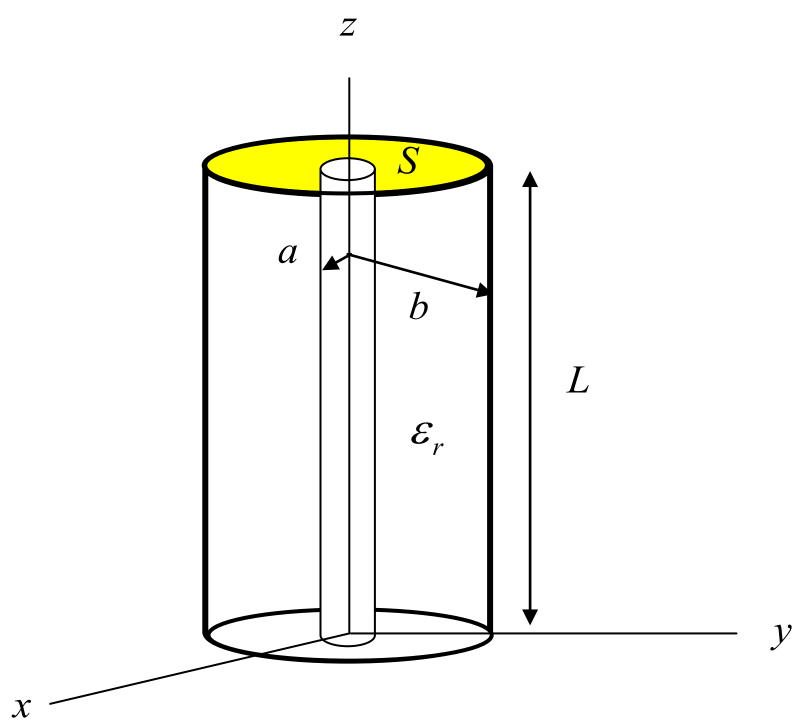
$$\underline{\mathscr{H}}(\rho,\phi,z,t) = \underline{\hat{\phi}}\left(\frac{1}{\rho}\right)\cos(kz)\cos\omega t \quad \text{[A/m]},$$

where $k = k_0\sqrt{\varepsilon_r}$ and $k_0 = \omega\sqrt{\mu_0\varepsilon_0}$.

a) Find the magnetic field vector in the phasor domain.

b) Find the electric field vector in the phasor domain.

c) Find the complex power going through the surface *S* in the downward sense. The surface *S* is an annular region that lies in the $z = L$ plane and is defined by $a < \rho < b$ and $0 < \phi < 2\pi$. Also find the time-average power in watts and the reactive power in vars going downward though the surface *S*.



In the phasor domain we have

$$\underline{H}(\rho,\phi,z)=\underline{\hat{\phi}}\left(\frac{1}{\rho}\right)\cos(kz) \quad \text{[A/m]}.$$

To find the electric field we use Amperes' law:

$$\nabla\times\underline{H}(\rho,\phi,z)=j\omega\varepsilon\underline{E}$$

so

$$\underline{E}=\frac{1}{j\omega\varepsilon_0\varepsilon_r}\nabla\times\underline{H}(\rho,\phi,z)=\frac{1}{j\omega\varepsilon_0\varepsilon_r}\nabla\times\underline{H}(\rho,\phi,z)$$

We have

$$\nabla\times\underline{H}=\underline{\hat{\rho}}\left(\frac{1}{\rho}\frac{\partial H_z}{\partial\phi}-\frac{\partial H_\phi}{\partial z}\right)+\underline{\hat{\phi}}\left(\frac{\partial H_\rho}{\partial z}-\frac{\partial H_z}{\partial\rho}\right)+\underline{\hat{z}}\frac{1}{\rho}\left(\frac{\partial(\rho H_\phi)}{\partial\rho}-\frac{\partial H_\rho}{\partial\phi}\right)$$

Keeping only the terms that survive, we have

$$\nabla\times\underline{H}=\underline{\hat{\rho}}\left(-\frac{\partial H_\phi}{\partial z}\right)+\underline{\hat{z}}\frac{1}{\rho}\left(\frac{\partial(\rho H_\phi)}{\partial\rho}\right)$$

so

$$\nabla\times\underline{H}=\underline{\hat{\rho}}\left(-\frac{\partial}{\partial z}\left(\left(\frac{1}{\rho}\right)\cos(kz)\right)\right)+\underline{\hat{z}}\frac{1}{\rho}\left(\frac{\partial\left(\cancel{\rho\left(\frac{1}{\rho}\right)}\cos(kz)\right)}{\partial\rho}\right)$$

or

$$\begin{aligned}\nabla\times\underline{H}&=\underline{\hat{\rho}}\left(-\frac{1}{\rho}\frac{\partial}{\partial z}(\cos(kz))\right)\\&=\underline{\hat{\rho}}\left(\frac{k}{\rho}\sin(kz)\right)\end{aligned}$$

Hence, we have

$$\underline{E}=\frac{1}{j\omega\varepsilon_0\varepsilon_r}\nabla\times\underline{H}(\rho,\phi,z)=\frac{1}{j\omega\varepsilon_0\varepsilon_r}\underline{\hat{\rho}}\left(\frac{k}{\rho}\sin(kz)\right)$$

or

$$\underline{E} = -j\underline{\hat{\rho}}\left(\frac{1}{\rho}\right)\frac{k}{\omega\varepsilon_0\varepsilon_r}\sin(kz) \quad [\text{V/m}].$$

The Poynting vector is

$$S = \frac{1}{2}\underline{E} \times \underline{H}^*.$$

Hence, we have

$$S = \frac{1}{2}\underline{\hat{z}}\left(-j\left(\frac{1}{\rho}\right)\frac{k}{\omega\varepsilon_0\varepsilon_r}\sin(kz)\right)\left(\left(\frac{1}{\rho}\right)\cos(kz)\right)^*.$$

Hence we have

$$S = -j\underline{\hat{z}}\frac{1}{2}\left(\frac{1}{\rho^2}\right)\left(\frac{k}{\omega\varepsilon_0\varepsilon_r}\right)(\sin(kz)\cos(kz)).$$

We then have

$$P_c = \int_S \underline{S}\cdot(-\underline{\hat{z}})\,dS = \int_0^{2\pi}\int_a^b -S_z\,\rho\,d\rho\,d\phi\,.$$

This gives us the complex power flowing down into the region as

$$P_c = j\frac{1}{2}\left(\frac{k}{\omega\varepsilon_0\varepsilon_r}\right)(\sin(kL)\cos(kL))(2\pi)\ln\left(\frac{b}{a}\right).$$

Hence,

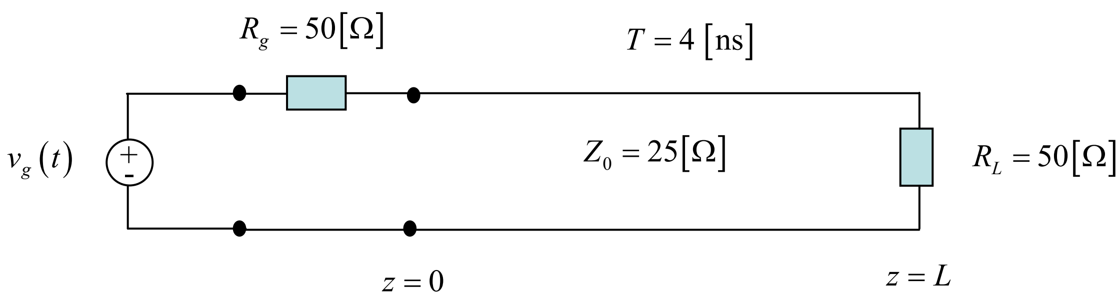
$$\text{Watts} = 0$$

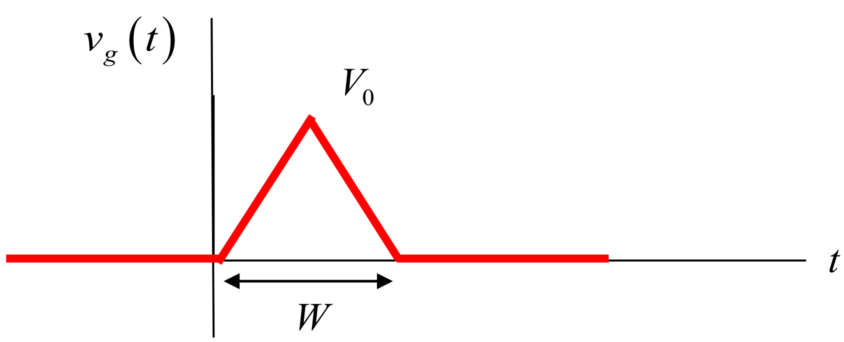
$$\text{Vars} = \frac{1}{2}\left(\frac{k}{\omega\varepsilon_0\varepsilon_r}\right)(\sin(kL)\cos(kL))(2\pi)\ln\left(\frac{b}{a}\right)$$

Problem 2 (35 pts)

A voltage source is applied at the left end of a transmission line as shown below. A plot of the generator voltage $v_g(t)$ is shown below. The pulse peak is $V_0 = 9\ [V]$ and the width of the pulse is $W = 2\ [\text{ns}]$.

Plot the voltage $v(t)$ at a point halfway down the line ($z = L/2$). Plot to a time of 10 [ns]. Use the graph on the next page to make your plot. Label all voltage values on your plot.





For this problem we use

$$v^{+}(z,t) = A v_g\left(t - z / c_d\right)$$

$$v^{-}(z,t) = \Gamma_L A v_g\left(t - L / c_d - (L - z) / c_d\right)$$

$$v^{++}(z,t) = \Gamma_g \Gamma_L A v_g\left(t - 2L / c_d - z / c_d\right)$$

$\vdots$

etc.

Note that the first two terms are sufficient, since the third wave does not reach the observation point until 10 [ns].

We then have

$$v^{+}(L/2,t) = \left(\frac{1}{3}\right) v_g\left(t - 0.5L / c_d\right)$$
$$v^{-}(L/2,t) = \left(\frac{1}{3}\right)\left(\frac{1}{3}\right) v_g\left(t - L / c_d - (0.5L) / c_d\right)$$

or

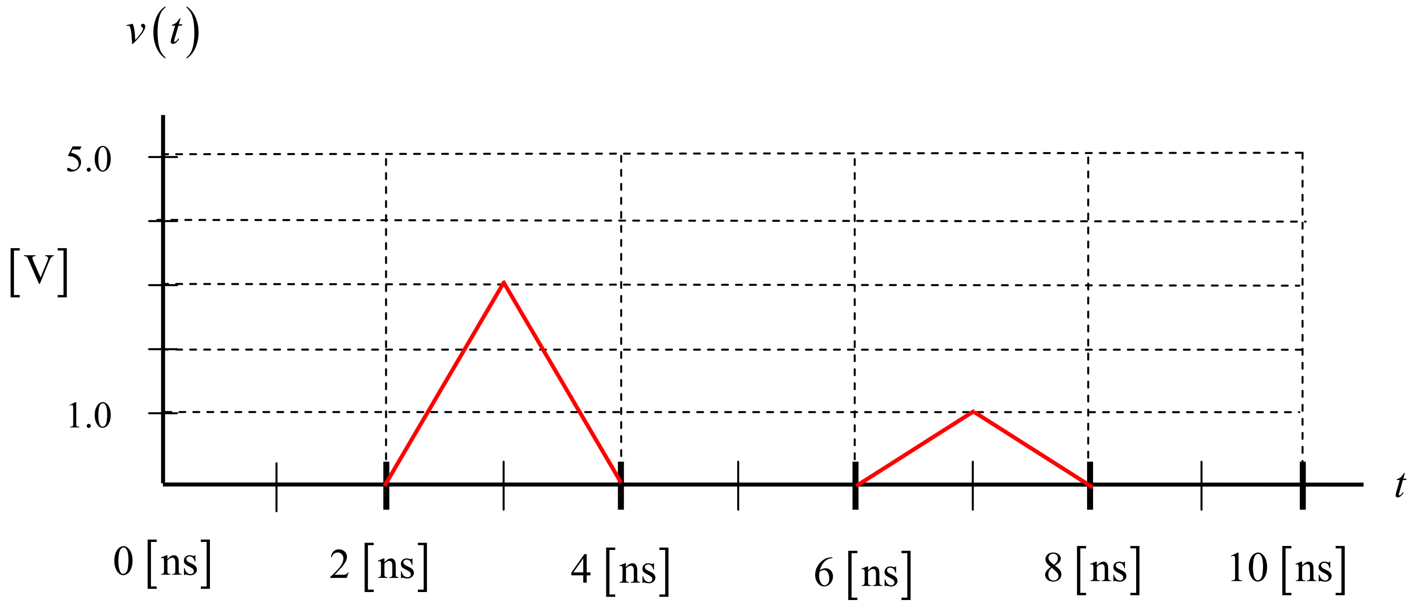
$$v^{+}(z,t) = \frac{1}{3} v_g\left(t - 2[\text{ns}]\right)$$

$$v^{-}(z,t) = \frac{1}{9} v_g\left(t - 6[\text{ns}]\right).$$

Hence,

$$v(L/2,t) = \frac{1}{3} v_g\left(t - 2[\text{ns}]\right) + \frac{1}{9} v_g\left(t - 6[\text{ns}]\right).$$

The plot is shown below.



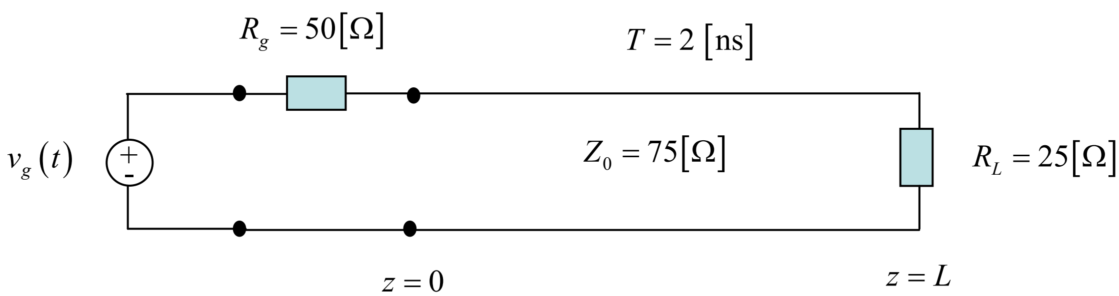
$v(t)$
5.0
[V]
1.0
t
0 [ns]
2 [ns]
4 [ns]
6 [ns]
8 [ns]
10 [ns]

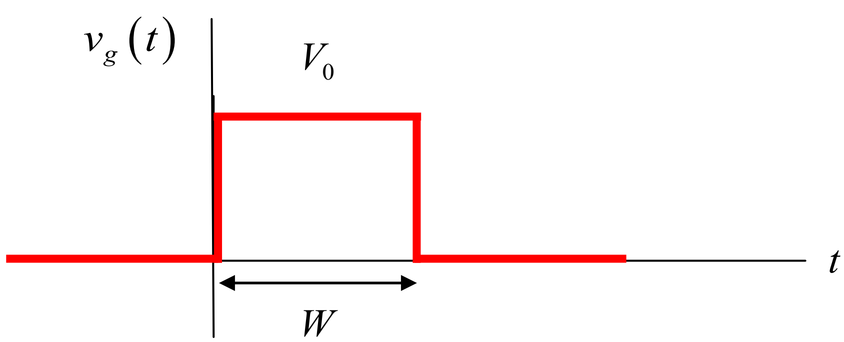
Problem 3 (35 pts)

A voltage source is applied at the left end of a transmission line as shown below. A plot of the generator voltage is shown below. The pulse width is $W = 2\,[\text{ns}]$ and the pulse voltage is $V_0 = 10\,[\text{V}]$.

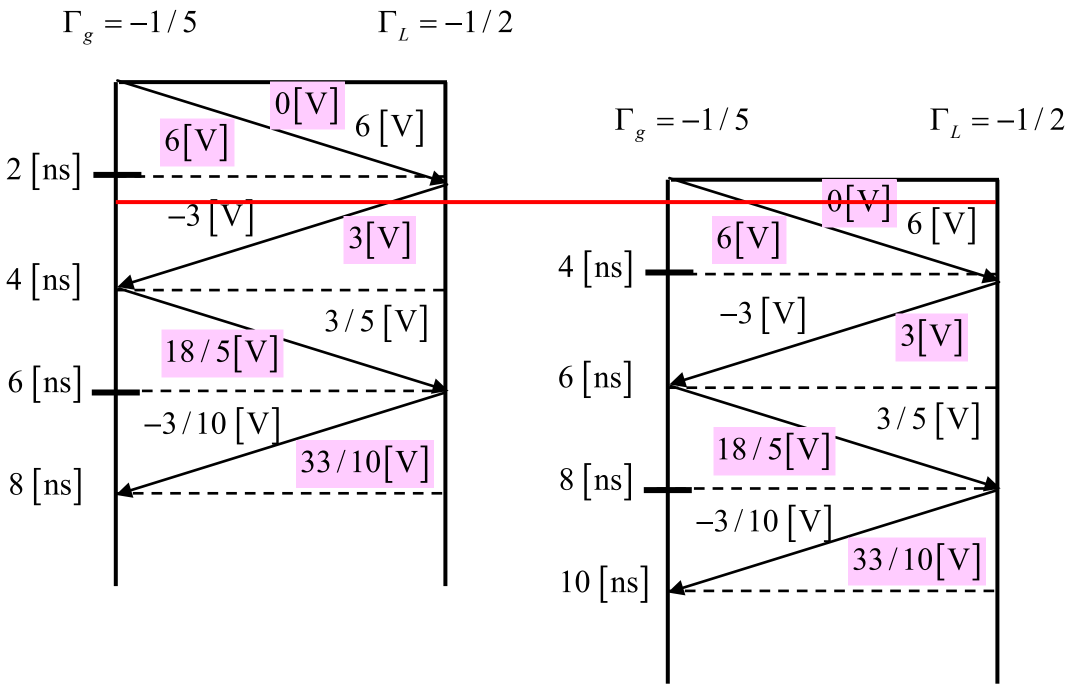
a) Construct a bounce diagram for this problem that extends to a time of 8 [ns]. (Make your bounce diagram on the next page.)

b) Make an accurate "snapshot" plot of the voltage $v(z)$ on the line at $t = 2.5\,[\text{ns}]$. Make your plot on the graph that is given below on the next page. Label all voltage values on your plot. Also indicate which directions the wavefronts are moving in.

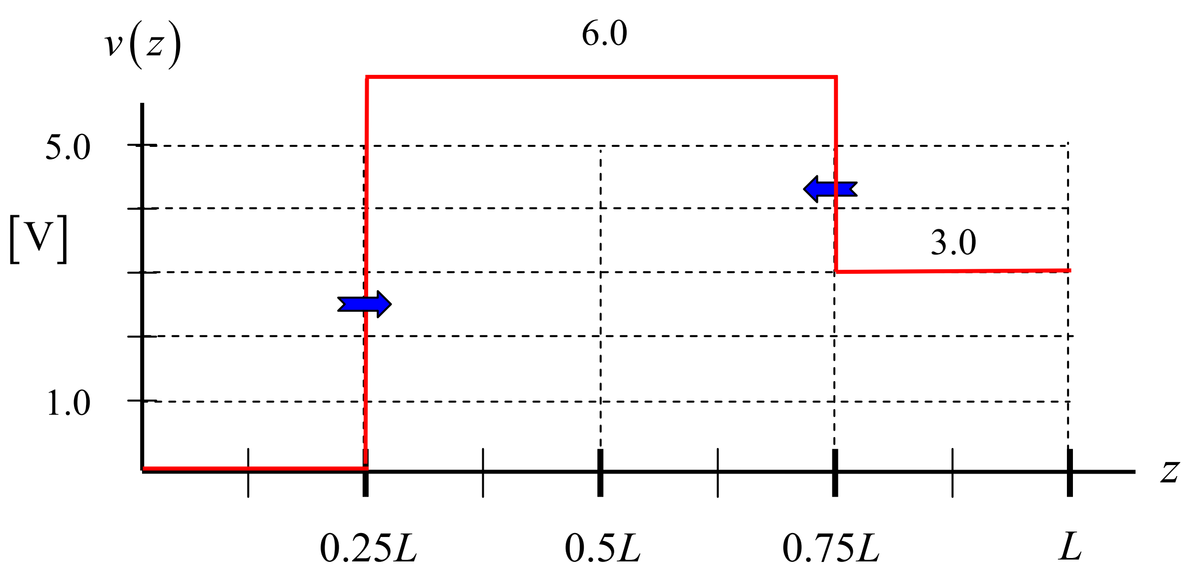




The bounce diagram is shown below.



The snapshot is shown below.



Problem 1 (30 pts)

An electric field in free space is described by

$$\underline{\mathscr{E}}(x,y,z,t) = \underline{\hat{z}}\cos(\omega t + \pi/4 + k_0 y) \quad \text{[V/m]},$$

where $k_0 = \omega\sqrt{\mu_0\varepsilon_0}$.

a) Find the electric field vector in the phasor domain.

b) Find the magnetic field vector in the phasor domain.

c) Find the complex power going through a surface S in the upward sense. The surface S is a square that is 2 meters × 2 meters, and the face of the square is perpendicular to the vector $\underline{V} = \left(\underline{\hat{x}}(1) + \underline{\hat{y}}(2) + \underline{\hat{z}}(3)\right)$.

Note: Please evaluate all constants that appear in your answers.

Solution

Part (a)

$$\underline{E}(x,y,z) = \underline{\hat{z}}\, e^{j\pi/4} e^{jk_0 y} \quad \text{[V/m]}$$

Part (b)

From Faraday's law (taking the curl of $\underline{E}$), we have

$$\underline{H}(x,y,z) = \underline{\hat{x}}\left(-\frac{1}{\eta_0}\right) e^{j\pi/4} e^{jk_0 y} \quad \text{[A/m]}$$

Part (c)

The complex Poynting vector is

$$\underline{S} = \frac{1}{2}\underline{E}\times\underline{H}^* = -\underline{\hat{y}}\frac{1}{2\eta_0}.$$

The unit normal to the surface is

$$\underline{\hat{n}} = \frac{1}{\sqrt{14}}\left(\underline{\hat{x}}(1) + \underline{\hat{y}}(2) + \underline{\hat{z}}(3)\right).$$

The power flowing through the surface is

$$P = \int_S \underline{S} \cdot \underline{\hat{n}}\, dS .$$

Because the Poynting vector is constant on the surface, we have

$$P = 4\left(\underline{S} \cdot \underline{\hat{n}}\right).$$

This gives us

$$P = 4\left(-\underline{\hat{y}}\frac{1}{2\eta_0}\right) \cdot \left(\frac{1}{\sqrt{14}}\left(\underline{\hat{x}}(1) + \underline{\hat{y}}(2) + \underline{\hat{z}}(3)\right)\right).$$

We then have

$$P = 4\left(-\frac{1}{2\eta_0}\frac{2}{\sqrt{14}}\right)$$

or

$$P = -\frac{4}{\sqrt{14}}\frac{1}{\eta_0} \ \left[\mathrm{VA}\right].$$

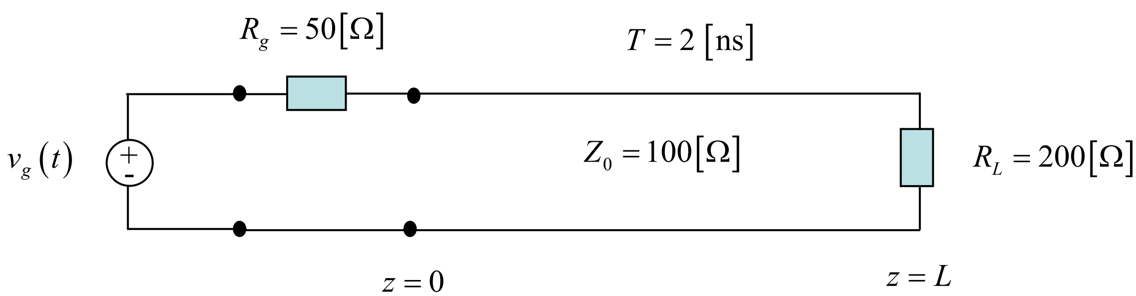
Note: Because the complex power is purely real, we can also say that this is the time-average power in Watts flowing through the surface.

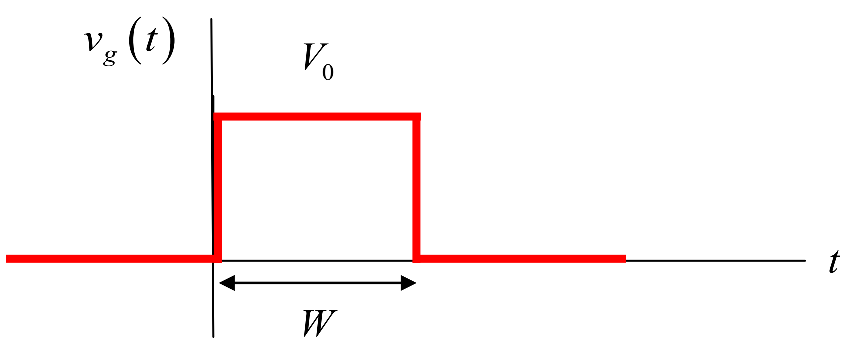
Problem 2 (35 pts)

A digital pulse of amplitude V_0 = 3.0 [V] and duration W = 1.5 [ns] is applied at the input to the transmission line circuit shown below.

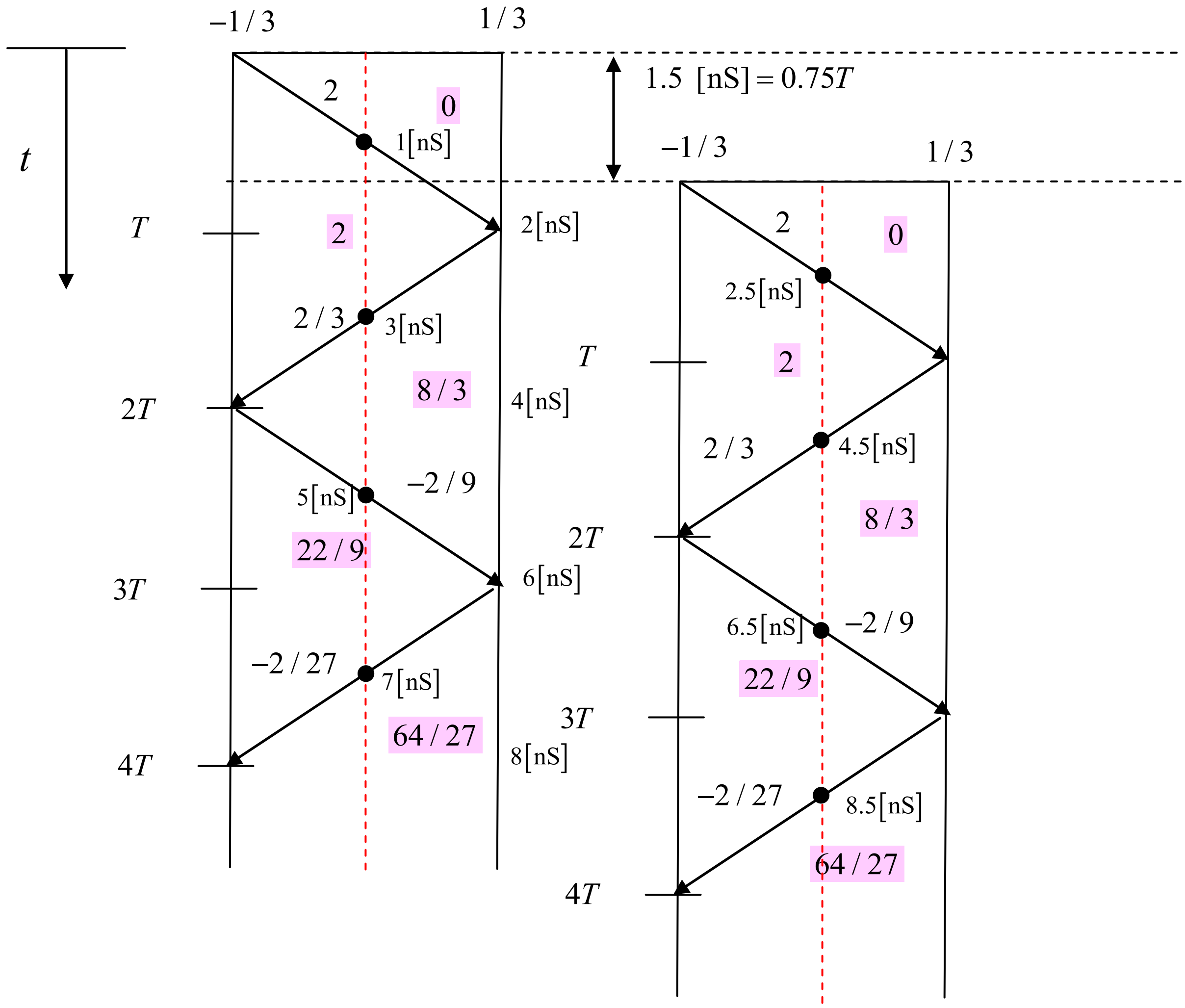
a) Construct a bounce diagram for this problem that extends to a time of 4T. (Make your bounce diagram on the next page.)

b) Make an accurate "oscilloscope trace" plot of the voltage $v(t)$ on the line at $z = L/2$. Make your plot on the graph that is given below. Label all voltage values on your plot, and label all times where the voltage changes. Plot to a time of 8 [ns].

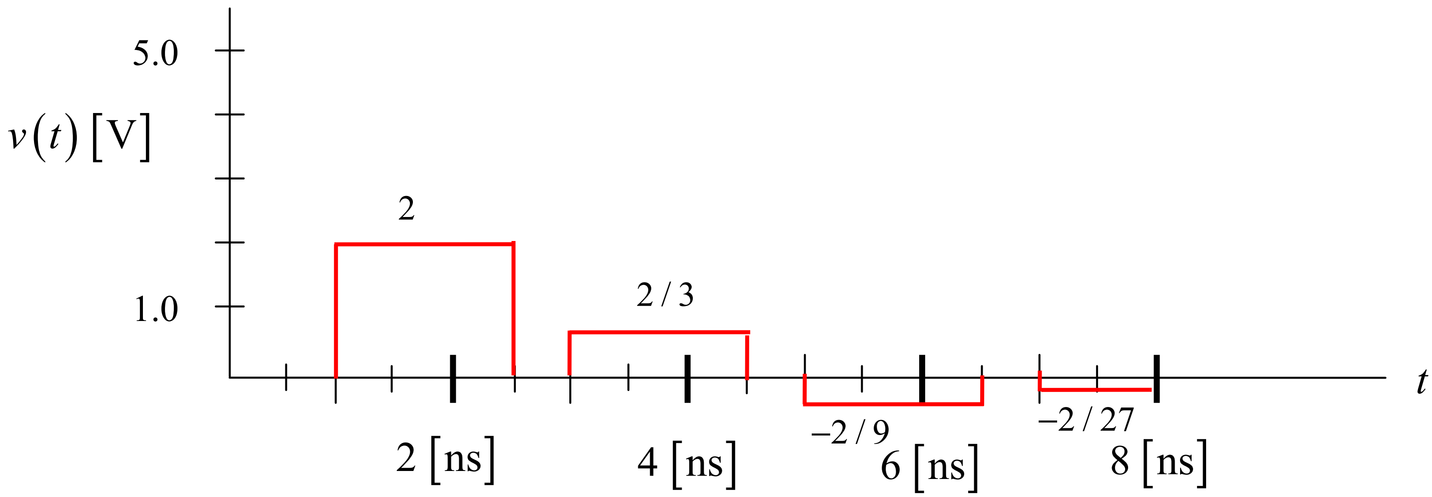




Make your bounce diagram here:



Make your plot here:

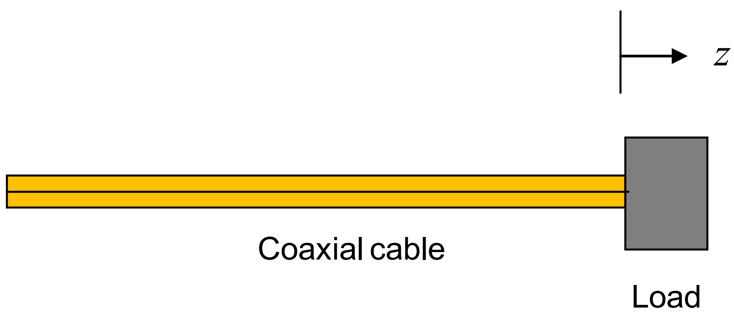


Problem 3 (35 pts.)

A coaxial cable transmission line has a characteristic impedance of 50 [Ω]. The relative permittivity of the (nonmagnetic) Teflon filling the line is 2.1. It is found that a voltage minimum of 2.0 [V] occurs on the line at z = -5 [cm] and a voltage maximum of 4.0 [V] occurs on the line at z = -15 [cm]. (This voltage maximum is the one that is the closest to the voltage minimum.)

What is the unknown load impedance at z = 0?

Note: This is not supposed to be a Smith chart problem, so please do not use a Smith chart.



Solution

We have

$\text{SWR} = 2$.

Next, use

$$\left|\Gamma_L\right| = \frac{\text{SWR-1}}{\text{SWR+1}}$$

to obtain

$$\left|\Gamma_L\right| = \frac{1}{3}.$$

For the phase of the load reflection coefficient, we have

$$\phi + 2\beta z = -\pi$$

so

$$\phi - 2\left(\frac{2\pi}{\lambda}\right)d = -\pi .$$

From the given data we have

$$d = -5[\text{cm}],$$

$$\lambda = \lambda_d = 4\left(-5-\left(-15\right)\right)[\text{cm}] = 40[\text{cm}].$$

Hence, we have

$$\phi = -\pi/2\ [\text{radians}].$$

Hence, we have

$$\Gamma_L = \frac{1}{3}e^{-j\pi/2} = -j/3.$$

The load impedance is then

$$Z_L = Z_0 \left(\frac{1+\Gamma_L}{1-\Gamma_L} \right).$$

This gives us

$$Z_L = 50 \left(\frac{1+(-j/3)}{1-(-j/3)} \right)$$

or

$$Z_L = 50 \left(\frac{1-j/3}{1+j/3} \right).$$

This gives us

$$Z_L = 40 - j30 \, [\Omega].$$

Problem 1 (30 pts)

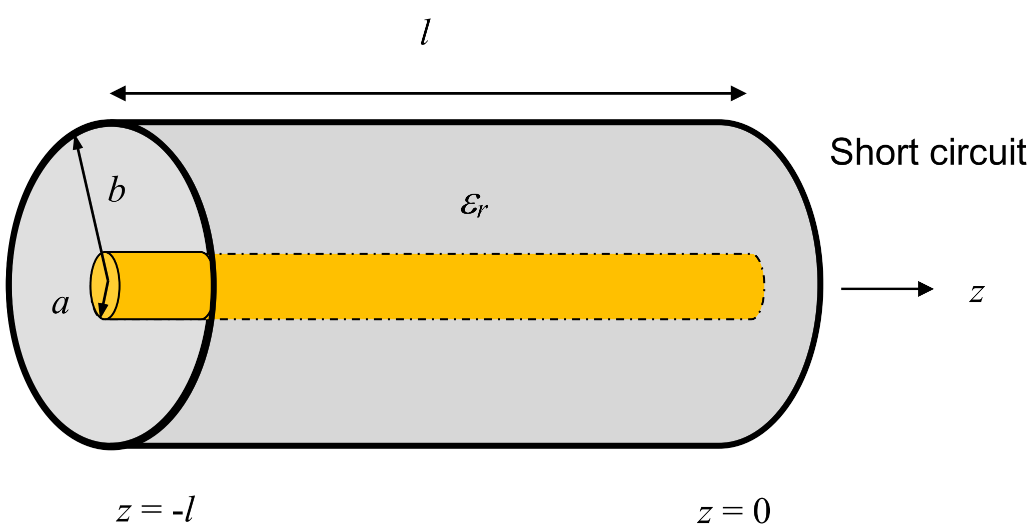
A lossless coaxial cable transmission line of length l is short circuited at the end (z = 0) with a short-circuit conducting plate. The electric field inside the coax ($a < \rho < b$) is given in the phasor domain by

$$\underline{E} = \underline{\hat{\rho}}\left(\frac{1}{\rho}\right)\left(\frac{-V_0}{\sin(kl)\ln(b/a)}\right)\sin(kz) \quad [\text{V/m}], \qquad -l < z < 0.$$

a) Find the magnetic field inside the coax (in the phasor domain).

b) Find the complex power entering into the coax.

c) Find the electric and magnetic fields inside the coax in the time domain.



Room for work

Part (a)

From Ampere's law, we have

$$\nabla\times\underline{E}=-j\omega\underline{B}.$$

Hence, we have

$$\underline{H}=-\frac{1}{j\omega\mu_0}\nabla\times\underline{E}.$$

Taking the curl in cylindrical coordinates, we have

$$\underline{H}=-\frac{1}{j\omega\mu_0}\left(\underline{\hat{\rho}}\left(\frac{1}{\rho}\frac{\partial E_z}{\partial\phi}-\frac{\partial E_\phi}{\partial z}\right)+\underline{\hat{\phi}}\left(\frac{\partial E_\rho}{\partial z}-\frac{\partial E_z}{\partial\rho}\right)+\underline{\hat{z}}\frac{1}{\rho}\left(\frac{\partial\left(\rho E_\phi\right)}{\partial\rho}-\frac{\partial E_\rho}{\partial\phi}\right)\right)$$

For the coax we then have

$$\underline{H}=-\frac{1}{j\omega\mu_0}\underline{\hat{\phi}}\left(\frac{\partial E_\rho}{\partial z}\right).$$

Hence,

$$\underline{H}=-\frac{1}{j\omega\mu_0}\underline{\hat{\phi}}\left(\frac{1}{\rho}\right)\left(\frac{-V_0k}{\sin\left(kl\right)\ln\left(b/a\right)}\right)\cos(kz)\quad[\text{A/m}].$$

Part (b)

The complex power entering into the coax is

$$P_c=\frac{1}{2}\int_0^{2\pi}\int_a^b E_\rho H_\phi^*\,\rho\,d\rho\,d\phi=\pi\int_a^b E_\rho H_\phi^*\,\rho\,d\rho.$$

This gives us

$$P_c=\pi\int_a^b\left(\frac{1}{\rho}\right)\left(\frac{-V_0}{\sin\left(kl\right)\ln\left(b/a\right)}\right)\sin\left(k\left(-l\right)\right)\left(-\frac{1}{j\omega\mu_0}\left(\frac{1}{\rho}\right)\left(\frac{-V_0k}{\sin\left(kl\right)\ln\left(b/a\right)}\right)\cos\left(k\left(-l\right)\right)\right)^*d\rho.$$

or

$$P_c = +j\pi\left(\frac{-V_0}{\sin(kl)\ln(b/a)}\right)^2\left(\frac{k}{\omega\mu_0}\right)\sin(kl)\cos(kl)\int_a^b\left(\frac{1}{\rho}\right)d\rho\,.$$

Hence, we have

$$P_c = j\pi\left(\frac{V_0}{\sin(kl)\ln(b/a)}\right)^2\left(\frac{1}{\eta_0}\right)\sin(kl)\cos(kl)\ln\left(\frac{b}{a}\right)\quad[\text{VARS}].$$

Note: There is purely reactive power entering the coax, since it is a lossless system that is terminated by a lossless (short-circuit) plate.

Part (c)

In the time domain we have

$$\underline{\mathscr{E}} = \underline{\hat{\rho}}\left(\frac{1}{\rho}\right)\left(\frac{-V_0}{\sin(kl)\ln(b/a)}\right)\sin(kz)\cos(\omega t)\quad[\text{V/m}].$$

The magnetic field is

$$\underline{\mathscr{H}} = -j\frac{1}{\omega\mu_0}\underline{\hat{\phi}}\left(\frac{1}{\rho}\right)\left(\frac{V_0 k}{\sin(kl)\ln(b/a)}\right)\cos(kz)\quad[\text{A/m}].$$

In the time domain this is

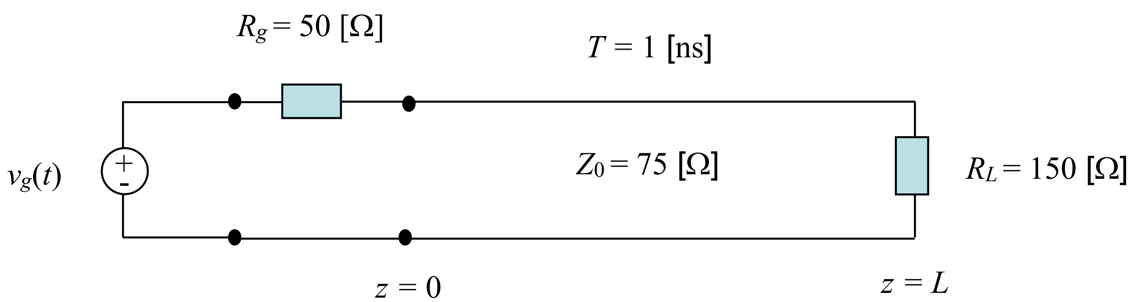
$$\underline{\mathscr{H}} = \frac{1}{\omega\mu_0}\underline{\hat{\phi}}\left(\frac{1}{\rho}\right)\left(\frac{V_0 k}{\sin(kl)\ln(b/a)}\right)\cos(kz)\sin(\omega t)\quad[\text{A/m}].$$

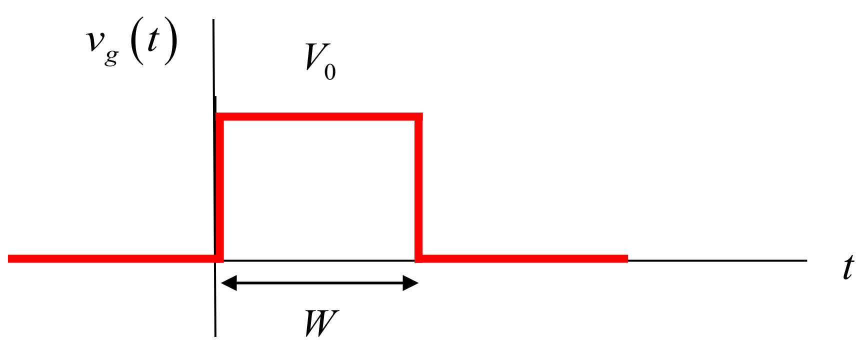
Problem 2 (30 pts)

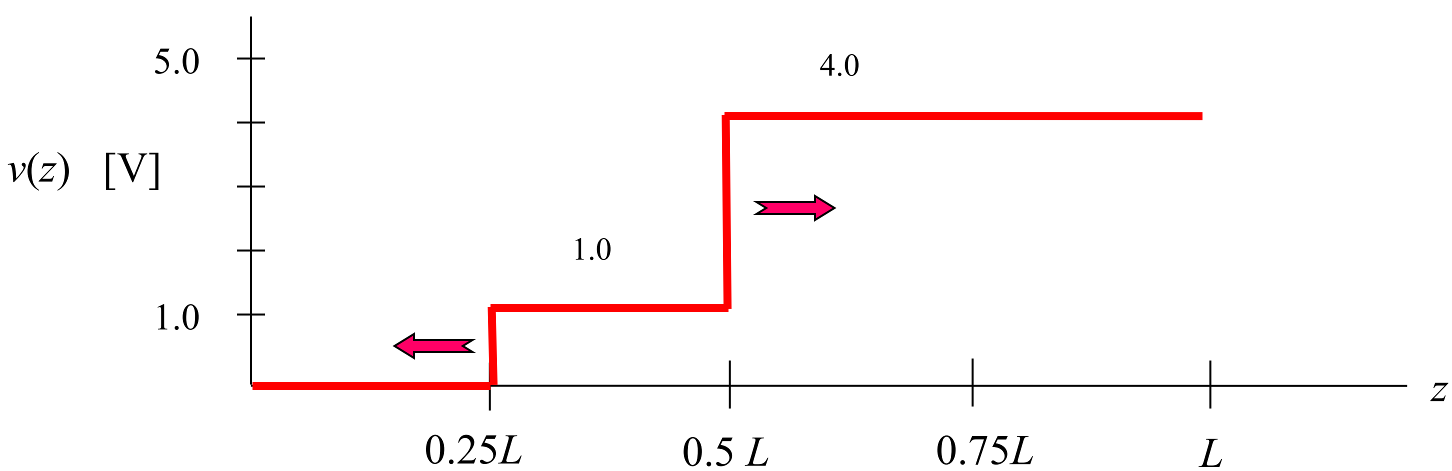
A digital pulse of amplitude $V_0 = 5.0$ [V] and duration $W = 1.25$ [ns] is applied at the input to the transmission line circuit shown below.

a) Construct a bounce diagram for this problem that extends to a time of $3T$. (Make your bounce diagram on the next page.)

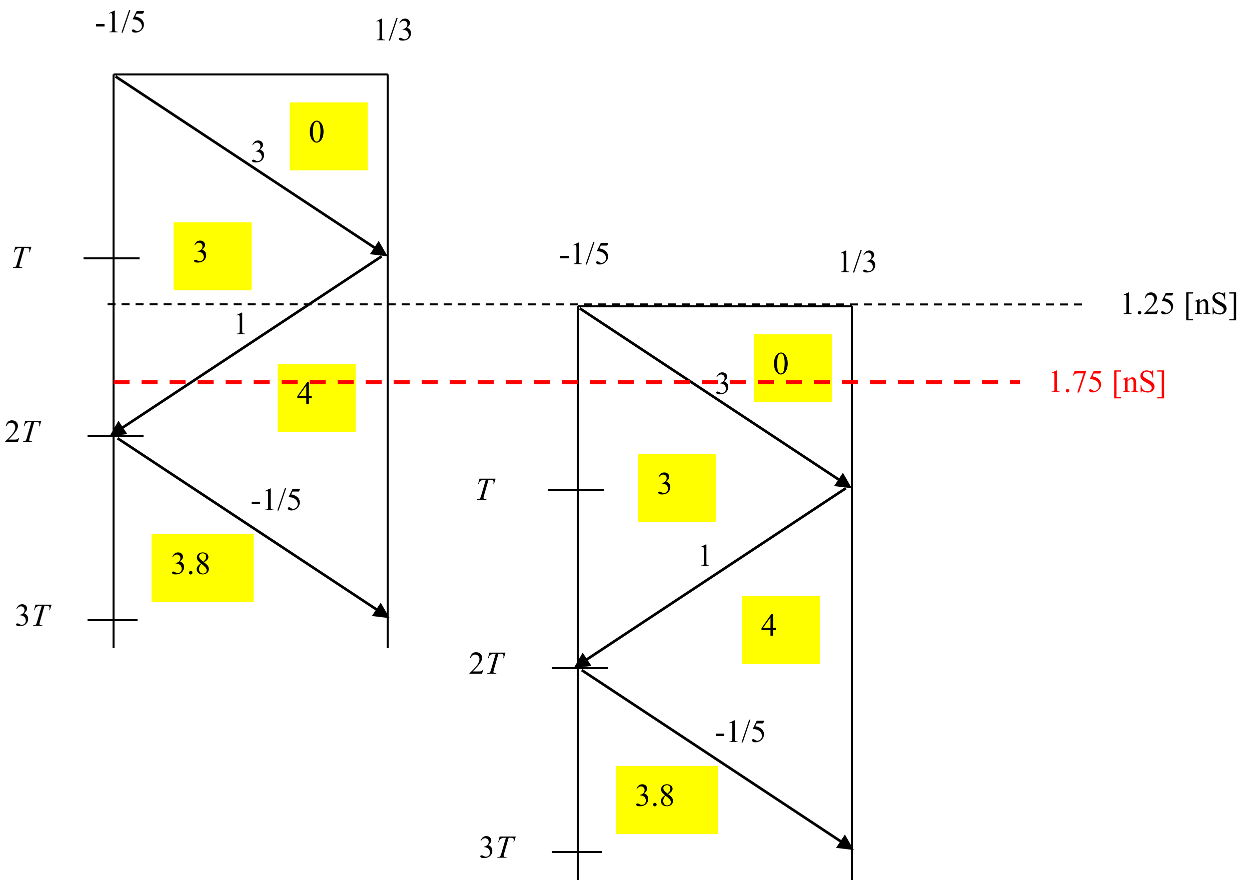
b) Make an accurate "snapshot" plot of the voltage on the line at $t = 1.75$ [ns]. Make your plot on the graph shown below. Label all voltage values on your plot. Also indicate with arrows in which direction all "wavefronts" (points of voltage discontinuity) are moving (to the left or to the right).







Room for work



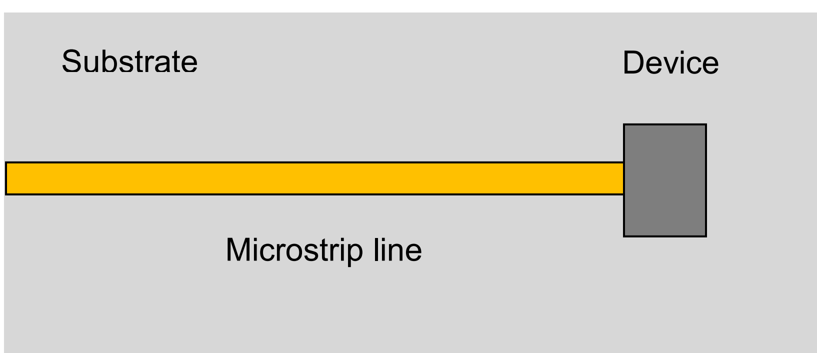
Problem 3 (40 pts.)

A transmission line is connected to a certain device on a printed circuit board, operating at 10.0 GHz (a top view is shown below). The device is located at z = 0. The microstrip line has a characteristic impedance of 50 [Ω]. The effective permittivity of the microstrip line is 1.5.

a) By probing the line, it is found that the voltage on the line has a maximum magnitude of 1.5 volts and a minimum magnitude of 0.5 volts. A voltage minimum occurs at a distance of 0.75 [cm] from the device. Determine the input impedance of the device. Do the calculation exactly (do not use the Smith chart).

b) Assume now that a new device is connected to the end of the line, which has an input impedance of $Z_{in} = 100+j100$ [Ω]. An open-circuited stub line having characteristic impedance of 50 [Ω] and an effective permittivity of 1.5 is added at a distance d from the load. Find the distance d and the length of the open-circuited stub line (in cm) to obtain a perfect match seen by an incoming wave that arrives from a generator on the left (not shown). Use the shortest distance d possible. Use the Smith chart to do all calculations. (A Smith chart is included at the end of this problem.)

c) As a continuation of part (b), what is the SWR on the main line between the device and the open-circuited stub? What is the SWR on the open-circuited stub line? What is the SWR on the main line to the left of the open-circuited stub? Do the calculations exactly (do not use the Smith chart).



Room for work

At 10.0 GHz we have

$$\lambda_0 = 2.9979 \ \ [\mathrm{cm}].$$

$$\lambda_{eff} = \frac{\lambda_0}{\sqrt{\varepsilon_r^{eff}}} = 2.4478 \ \ [\mathrm{cm}]$$

Part (a)

We have

$$Z_{in} = Z_0 \left(\frac{1+\Gamma_L}{1-\Gamma_L} \right)$$

with

$$\Gamma_L = \left|\Gamma_L\right| e^{j\phi}$$

and

$$\left|\Gamma_L\right| = \frac{\mathrm{SWR}-1}{\mathrm{SWR}+1}$$

$$\phi + 2\beta z_{min} = \pi + n\pi \ .$$

We thus have

$$\left|\Gamma_L\right| = \frac{3-1}{3+1} = \frac{1}{2}.$$

Also, we have

$$\phi + 2\left(2\pi \frac{z_{min}}{\lambda_{eff}} \right) = \pi + n\pi$$

or

$$\phi + 2\left(2\pi \frac{z_{min}}{\lambda_{eff}} \right) = \pi + n\pi \ .$$

This gives us

$$\phi = 0.70875 \quad [\text{rad}].$$

Hence

$$\Gamma_L = 0.5\, e^{j(0.70875)}.$$

This gives us

$$Z_{in} = 50\left(\frac{1+0.5\, e^{j(0.70875)}}{1-0.5\, e^{j(0.70875)}}\right).$$

The final result is

$$Z_{in} = 76.402 + j(66.305) \;\; [\Omega].$$

Part (b)

The Smith chart is omitted here. The solution is

$$d = 0.215\lambda_{eff} = 0.526 \; [\text{cm}]$$

$$l = 0.34\lambda_{eff} = 0.278\lambda_0 = 0.832[\text{cm}]$$

Part (c)

On the main line between the device and the stub, we have

$$\Gamma_L = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} = 0.538 + j(0.308)$$

Using

$$\text{SWR} = \frac{1+|\Gamma_L|}{1-|\Gamma_L|},$$

we have

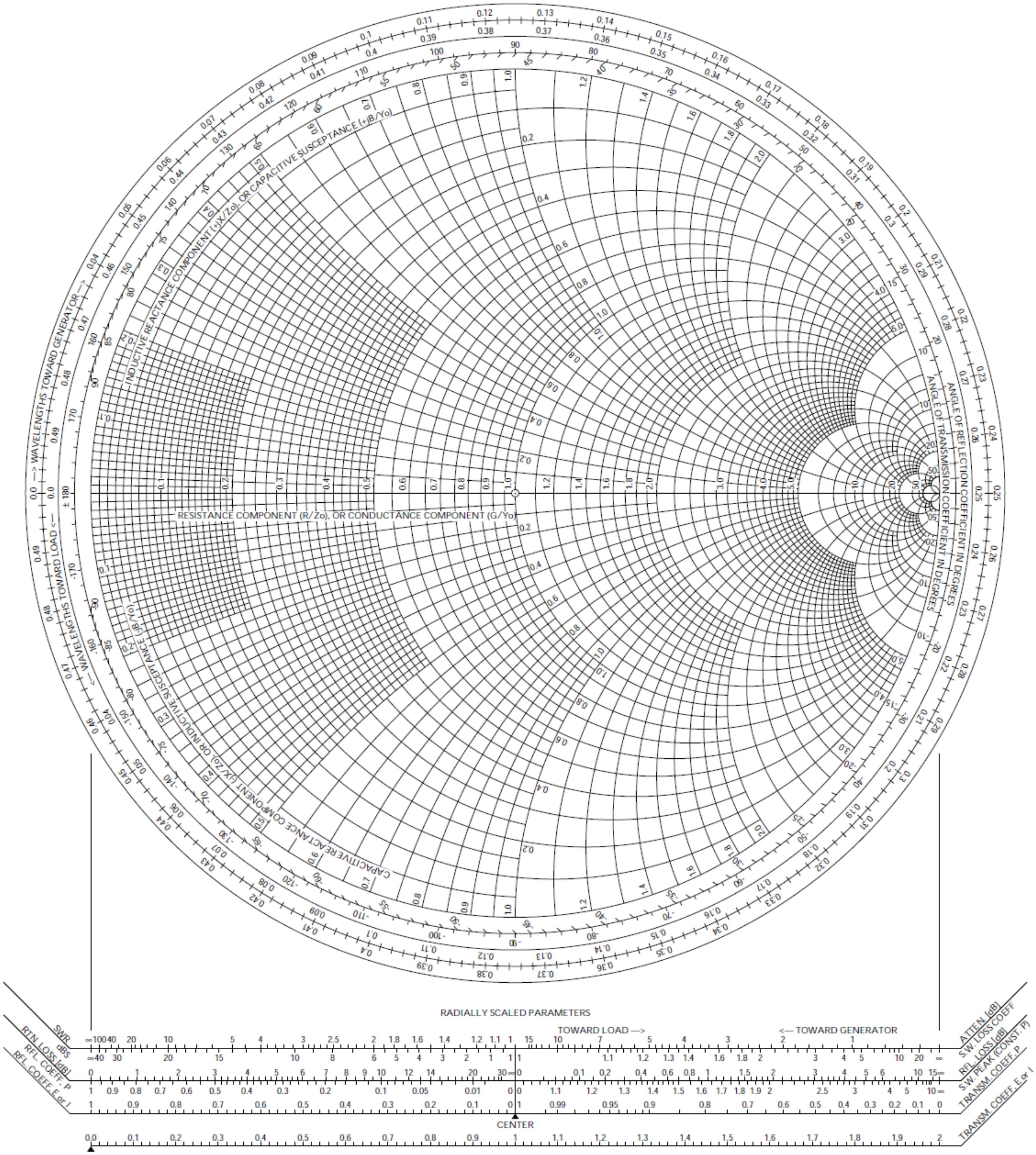
$\mathrm{SWR} = 4.26$.

On the open-circuited stub we have

$\mathrm{SWR} = \infty$.

On the main line to the left of the stub we have a perfect match, so we have

$\mathrm{SWR} = 1.0$.



Problem 1 (25 pts)

A lossy coaxial cable is shown below. The coax is filled with a lossy dielectric. The fields inside the coax are given (in cylindrical coordinates) by

$$\underline{E}(\rho,z)=\underline{\hat{\rho}}\left(\frac{1}{\rho}\right)e^{-j\beta z}e^{-\alpha z} \quad \text{[V/m]},$$

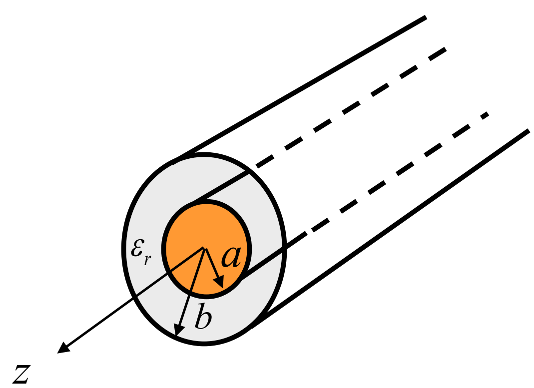
$$\underline{H}(\rho,z)=\underline{\hat{\phi}}\frac{1}{\eta}\left(\frac{1}{\rho}\right)e^{-j\beta z}e^{-\alpha z} \quad \text{[A/m]},$$

where

$$\eta=|\eta|e^{j\phi}$$

is a complex number written in polar form (this is the complex intrinsic impedance of the dielectric material).

Find the complex power that is flowing in the positive z direction at any point z on the coax.



ROOM FOR WORK

Solution

$$\underline{S} = \frac{1}{2}\underline{E}\times\underline{H}^*$$

$$\begin{aligned}
\underline{S} &= \frac{1}{2}\left(\hat{\underline{\rho}}\left(\frac{1}{\rho}\right)e^{-j\beta z}e^{-\alpha z}\right)\times\left(\hat{\underline{\phi}}\frac{1}{\eta}\left(\frac{1}{\rho}\right)e^{-j\beta z}e^{-\alpha z}\right)^* \\
&= \frac{1}{2}\left(\hat{\underline{\rho}}\times\hat{\underline{\phi}}\right)\left(\frac{1}{\rho^2}\right)\left(e^{-j\beta z}e^{-\alpha z}\right)\left(e^{-j\beta z}e^{-\alpha z}\right)^*\frac{1}{\eta^*} \\
&= \frac{1}{2}\hat{\underline{z}}\left(\frac{1}{\rho^2}\right)\left(e^{-j\beta z}e^{-\alpha z}\right)\left(e^{+j\beta z}e^{-\alpha z}\right)\frac{1}{\eta^*} \\
&= \frac{1}{2}\hat{\underline{z}}\left(\frac{1}{\rho^2}\right)\left(e^{-2\alpha z}\right)\frac{1}{\eta^*} \\
&= \frac{1}{2}\hat{\underline{z}}\left(\frac{1}{\rho^2}\right)\left(e^{-2\alpha z}\right)\frac{1}{|\eta|e^{-j\phi}} \\
&= \frac{1}{2}\hat{\underline{z}}\left(\frac{1}{\rho^2}\right)\left(e^{-2\alpha z}\right)\frac{1}{|\eta|}e^{+j\phi}.
\end{aligned}$$

Hence,

$$\underline{S} = \frac{1}{2}\hat{\underline{z}}\left(\frac{1}{\rho^2}\right)\left(e^{-2\alpha z}\right)\frac{1}{|\eta|}e^{+j\phi}\ \left[\mathrm{VA/m^2}\right].$$

We then have

$$\begin{aligned}
P_c &= \int_S \underline{S}\cdot\hat{\underline{z}}\,dS \\
&= \int_S S_z dS \\
&= \int_0^{2\pi}\int_a^b S_z\,\rho d\rho d\phi \\
&= \int_0^{2\pi}\int_a^b \frac{1}{2}\left(\frac{1}{\rho^2}\right)\left(e^{-2\alpha z}\right)\frac{1}{|\eta|}e^{+j\phi}\,\rho d\rho d\phi \\
&= \frac{1}{2}\left(e^{-2\alpha z}\right)\frac{1}{|\eta|}e^{+j\phi}\int_0^{2\pi}\int_a^b \frac{1}{\rho}d\rho d\phi.
\end{aligned}$$

Hence,

$$\begin{aligned} P_c &= \frac{1}{2}\left(e^{-2\alpha z}\right)\frac{1}{|\eta|}e^{+j\phi}\int_a^b \frac{1}{\rho}\,d\rho\int_0^{2\pi} d\phi \\ &= \frac{1}{2}\left(e^{-2\alpha z}\right)\frac{1}{|\eta|}e^{+j\phi}\left(\ln\left(\frac{b}{a}\right)\right)(2\pi). \end{aligned}$$

We then have

$$P_c = \pi\left(e^{-2\alpha z}\right)\frac{1}{|\eta|}\ln\left(\frac{b}{a}\right)(\cos\phi + j\sin\phi).$$

We then also have

$$\operatorname{Re} P_c = \pi\left(e^{-2\alpha z}\right)\frac{1}{|\eta|}\ln\left(\frac{b}{a}\right)\cos\phi\ [\mathrm{W}].$$

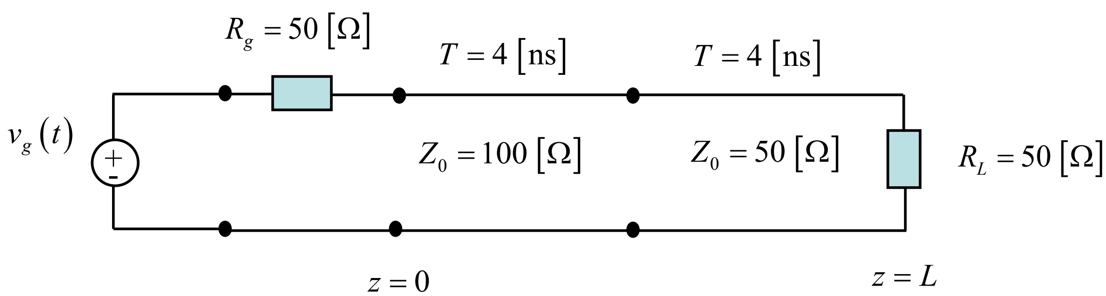
$$\operatorname{Im} P_c = \pi\left(e^{-2\alpha z}\right)\frac{1}{|\eta|}\ln\left(\frac{b}{a}\right)\sin\phi\ [\mathrm{Vars}].$$

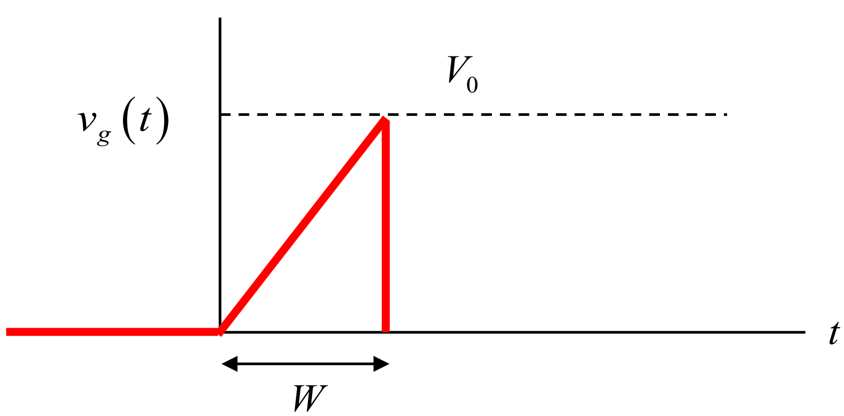
Problem 2 (25 pts)

A voltage source is applied at the left end of a transmission line as shown below. The transmission line meets a second transmission line, which is then terminated by a load.

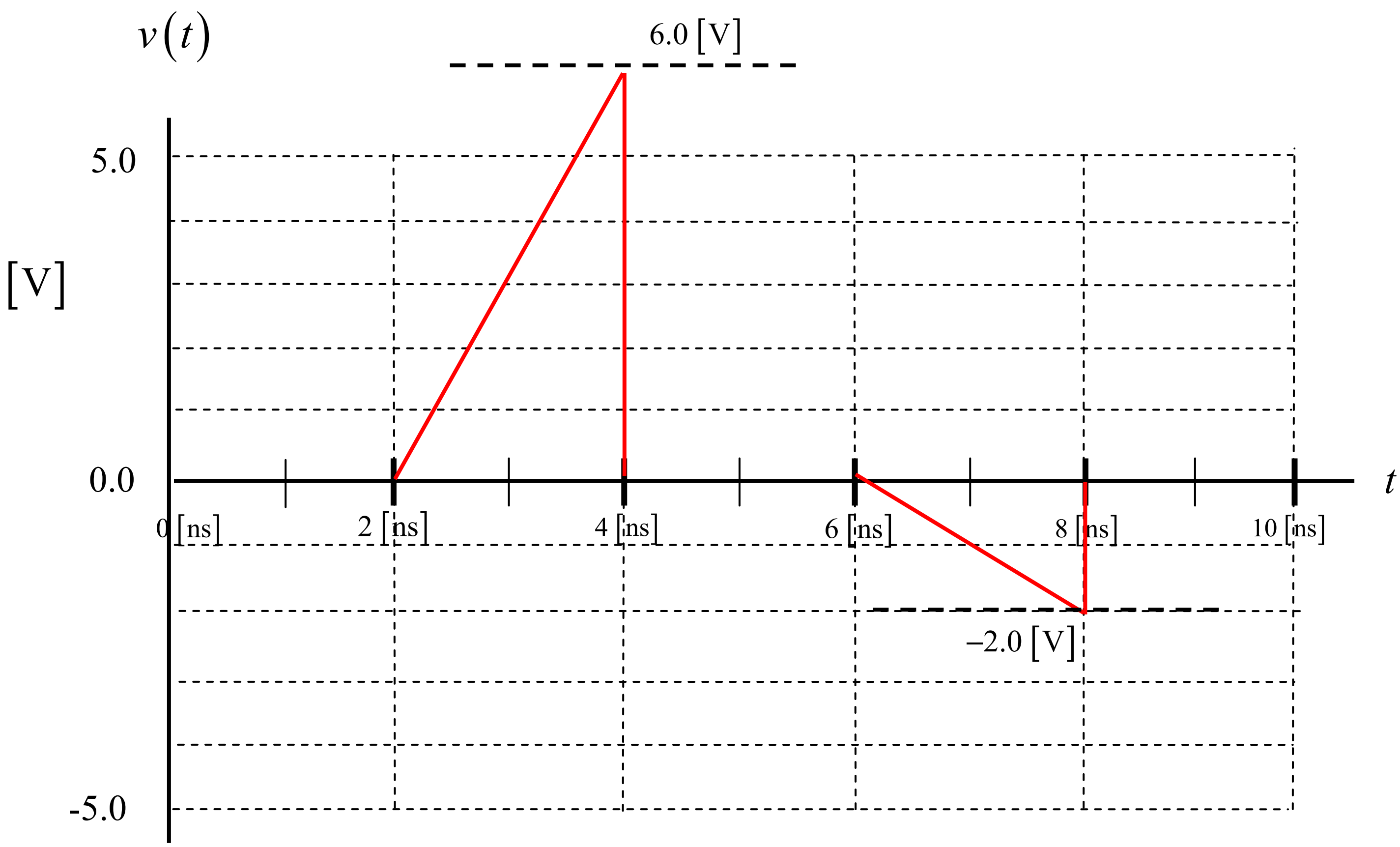
A plot of the generator voltage $v_g(t)$ is shown below. The pulse peak is $V_0 = 9\ [\mathrm{V}]$ and the width of the pulse is $W = 2\ [\mathrm{ns}]$.

Plot the voltage $v(t)$ measured by an oscilloscope that is connected to the first (left) line at a point halfway down the first line (halfway between the generator and the junction). Plot to a time of 10 [ns]. Use the graph on the next page to make your plot. Label all voltage values on your plot. (Please show your work on the page that is after the page with the plot.)





Make your plot here:



Solution

$$A = \frac{2}{3};\quad \Gamma_L = -\frac{1}{3};\quad \Gamma_g = -\frac{1}{3}$$

$$v(t) = Av_g\left(t-2[\text{nS}]\right) + A\Gamma_L v_g\left(t-6[\text{nS}]\right) + A\Gamma_L\Gamma_g v_g\left(t-10[\text{nS}]\right) + \ldots\ [\text{V}]$$

Keeping the first two terms, we have

$$v(t) = \left(\frac{2}{3}\right)v_g\left(t-2[\text{nS}]\right) + \left(\frac{2}{3}\right)\left(-\frac{1}{3}\right)v_g\left(t-6[\text{nS}]\right) + \ldots\ [\text{V}]$$

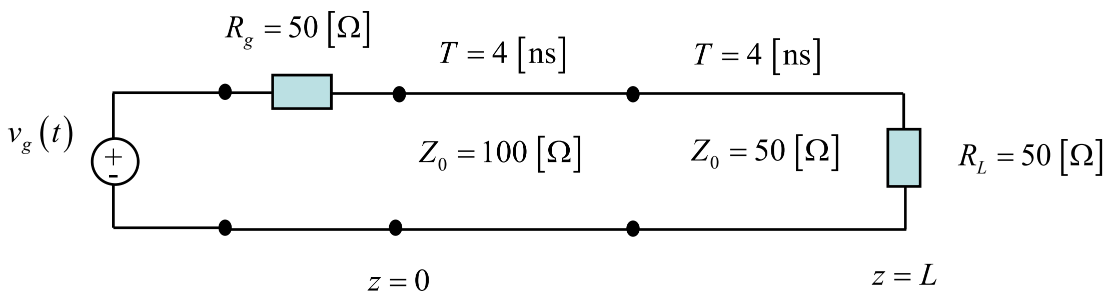
Problem 3 (25 pts)

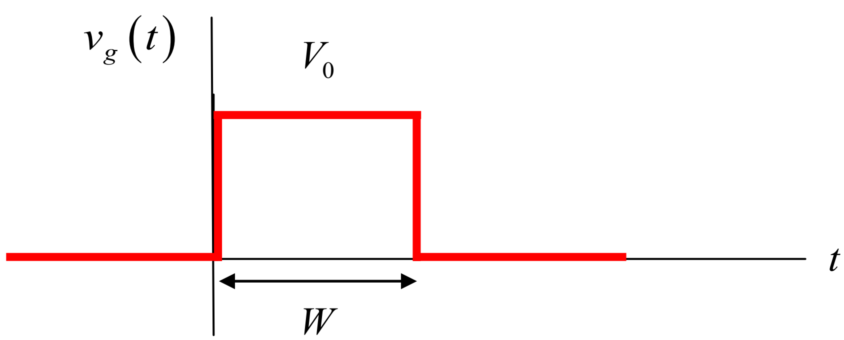
A voltage source is applied at the left end of a transmission line as shown below. The transmission line meets a second transmission line, which is then terminated by a load.

A plot of the generator voltage $v_g(t)$ is shown below. The pulse peak is $V_0 = 9\,[\mathrm{V}]$ and the width of the pulse is $W = 8\,[\mathrm{ns}]$.

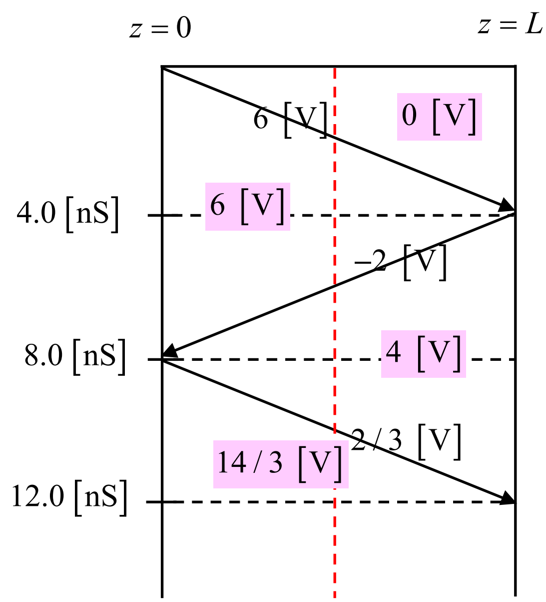
(a) Make a bounce diagram for this problem, for the left line. Plot to a time of 12 [ns]. Put your bounce diagram on the next page.

(b) Plot the voltage $v(t)$ measured by an oscilloscope that is connected to the first (left) line at a point halfway down the first line (halfway between the generator and the junction). Plot to a time of 10 [ns]. Use the graph on the next page to make your plot. Label all voltage values on your plot. Please show your work on the page that is after the page with your plot.

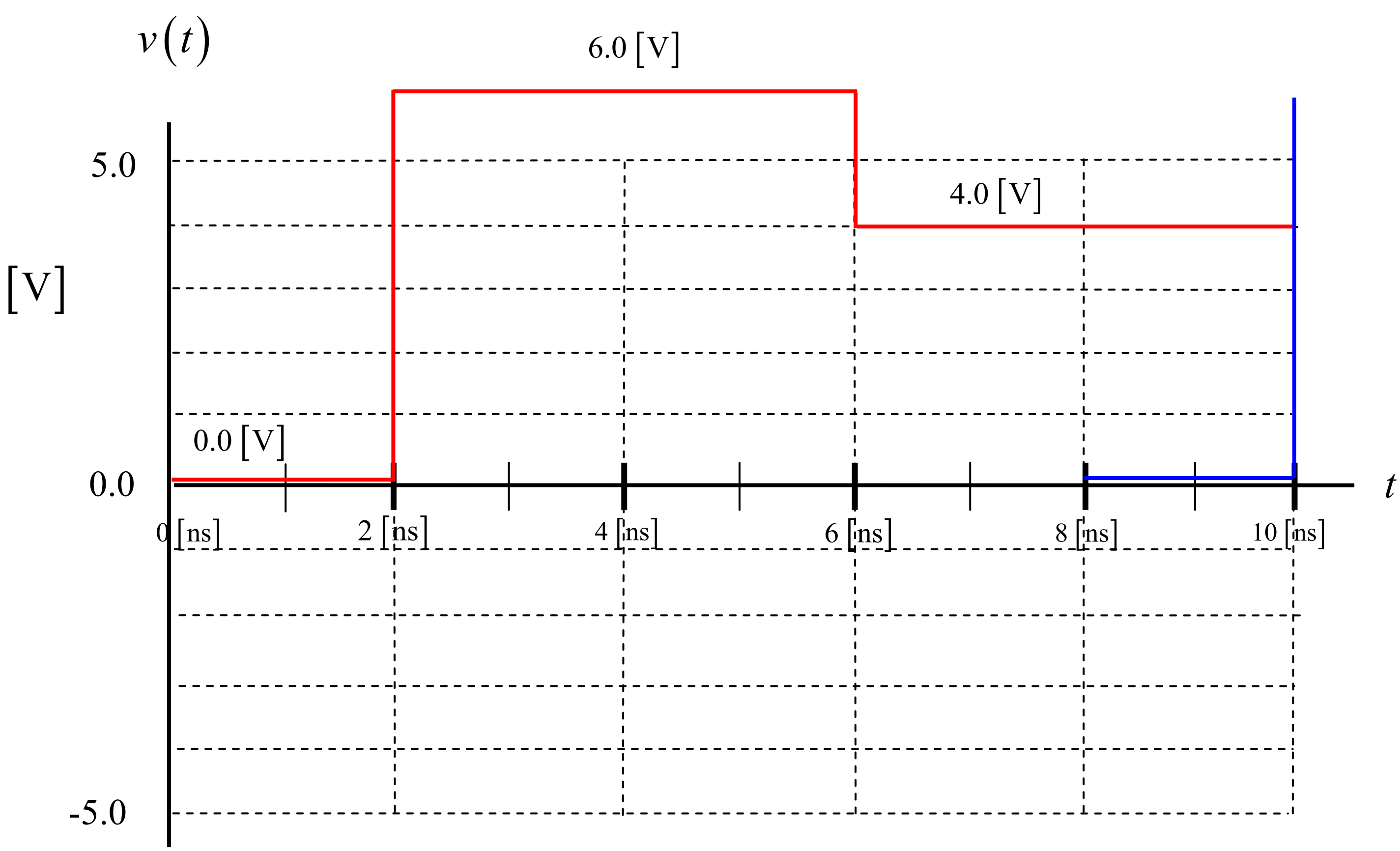




Make your bounce diagram here:



Make your plot here:



Solution

The oscilloscope trace from the step function of amplitude 6 [V] is shown in red on the plot. The oscilloscope trace due to the delayed step function of amplitude 6 [V] is shown in blue on the plot. As can be seen, when we subtract the blue curve from the red curve, the blue curve does not have any effect when we plot up to 10 [nS]. Hence, the red plot is also the final oscilloscope trace.

Problem 4 (25 pts)

At a frequency of 1.0 GHz, a coaxial cable has the following parameters:

$R = 4.5$ [Ω/m]

$L = 4.0\times10^{-7}$ [H/m]

$G = 5.0\times10^{-4}$ [S/m]

$C = 7.0\times10^{-11}$ [F/m]

Find the maximum length of the coaxial cable that we can have, if we wish to keep the loss less than 10 dB at 1.0 GHz.

Solution

We have

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}\,.$$

At 1.0 GHz this gives us

$$\gamma = 0.048663 + j33.247493\ [1/\mathrm{m}]\,.$$

Hence,

$$\alpha = 0.048663\ [\mathrm{np/m}]\,.$$

Therefore,

$$\mathrm{Att}_{\mathrm{dB/m}} = 0.048663(8.686)\ [\mathrm{dB/m}]\,.$$

We then have for the length L of the coaxial cable

$$(\mathrm{Att}_{\mathrm{dB/m}})L = 10\ [\mathrm{dB}]\,.$$

We thus have

$$L = \frac{10}{\mathrm{Att}_{\mathrm{dB/m}}}\,.$$

This gives us

$$L = \frac{10}{0.048663(8.686)}$$

so

$$L = 23.6582\ [\mathrm{m}].$$